

IP Addressing & Networking (Easy Learning Format)

IPv4 Address

- IPv4 is a 32-bit decimal address written as A.B.C.D
- Example: **192.168.10.1**

Binary Representation

192.168.10.1 → 11000000.10101000.00001010.00000001

Octet Range

- Minimum: 00000000 = 0
- Maximum: 11111111 = 255

IP Classes

- Class A → Range: 0–127 | Subnet Mask: 255.0.0.0 | CIDR: /8
- Class B → Range: 128–191 | Subnet Mask: 255.255.0.0 | CIDR: /16
- Class C → Range: 192–223 | Subnet Mask: 255.255.255.0 | CIDR: /24

Total Networks

- Class A: 126 valid networks
- Class B: 16,384 networks
- Class C: 2,097,152 networks

Total Hosts

- Class A: 16,777,214 hosts
- Class B: 65,534 hosts
- Class C: 254 hosts